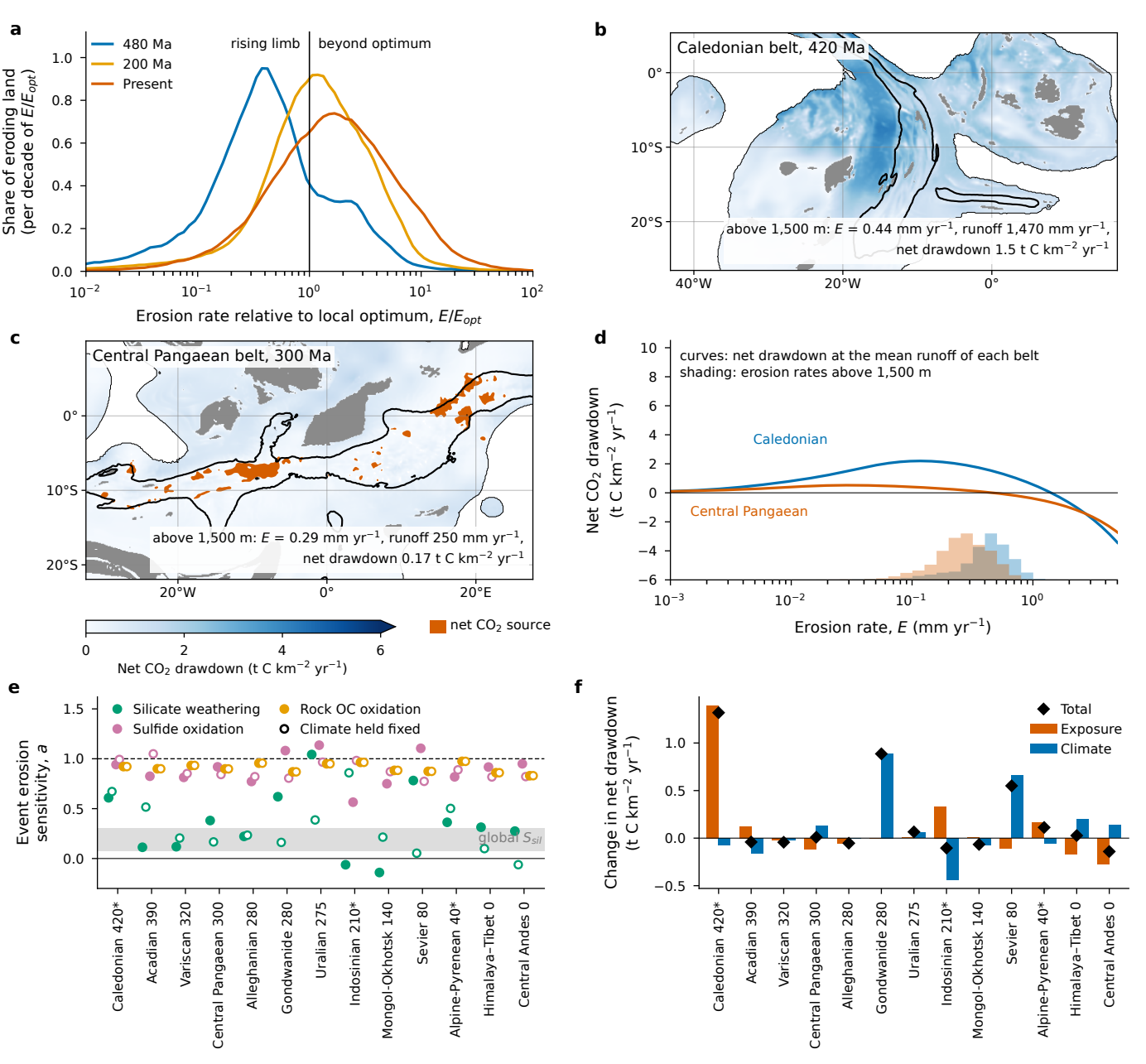

a
480 Ma
200 Ma
Present
rising limb
beyond optimum
Share of eroding land (per decade of E/E_{opt})
Erosion rate relative to local optimum, E/E_{opt}
b
Caledonian belt, 420 Ma
above 1,500 m: E = 0.44 mm yr^{-1}, runoff 1,470 mm yr^{-1}, net drawdown 1.5 t C km^{-2} yr^{-1}
c
Central Pangaean belt, 300 Ma
above 1,500 m: E = 0.29 mm yr^{-1}, runoff 250 mm yr^{-1}, net drawdown 0.17 t C km^{-2} yr^{-1}
net CO$_2$ source
Net CO$_2$ drawdown (t C km^{-2} yr^{-1})
d
curves: net drawdown at the mean runoff of each belt
shading: erosion rates above 1,500 m
Caledonian
Central Pangaean
Net CO$_2$ drawdown (t C km^{-2} yr^{-1})
Erosion rate, E (mm yr^{-1})
e
Silicate weathering
Sulfide oxidation
Rock OC oxidation
Climate held fixed
global S_{sil}
Event erosion sensitivity, a
Caledonian 420*
Acadian 390
Variscan 320
Central Pangaean 300
Alleghanian 280
Gondwanide 280
Uralian 275
Indosinian 210*
Mongol-Okhotsk 140
Sevier 80
Alpine-Pyrenean 40*
Himalaya–Tibet 0
Central Andes 0
f
Total
Exposure
Climate
Change in net drawdown (t C km^{-2} yr^{-1})